FR. CONCEICAO RODRIGUES COLLEGE OF ENGINEERING

Department of Computer Engineering
Academic Year 2025-26

Rubrics for Lab Experiments

Class : B.*E. Computer* **Subject Name :*BDA Lab Engineering***
Semester : VII **Subject Code :CSL702**

Practical No:	6
Title:	To analyze and gain insights into the social network dynamics of an online community using Neo4j, with a focus on understanding user interactions and community structure.
Date of Performance:	18/9/2025
Roll No:	9913
Name of the Student:	Mark Lopes

Evaluation:

Performance Indicator	Below average	Average	Good	Excellent	Marks
On time Submission (2)	Not submitted (0)	Submitted after deadline (1)	Early or on time submission(2)	---	
Test cases and output (4)	Incorr ect output (1)	The expected output is verified only a for few test cases (2)	The expected output is Verified for all test cases but is not presentable (3)	Expected output is obtained for all test cases. Presentable and easy to follow (4)	
Coding efficiency (2)	The code is not structured at all (0)	The code is structured but not efficient (1)	The code is Structured and efficient. (2)	-	

Knowledge(2)	Basic concepts not clear (0)	Understood the basic concepts (1)	Could explain the concept with suitable example (1.5)	Could relate the theory with real world application(2)	
Total					

Signature of the Teacher:

<u>Experiment 6</u>

Aim: To analyze and gain insights into the social network dynamics of an online community using Neo4j, with a focus on understanding user interactions and community structure.

Objective:

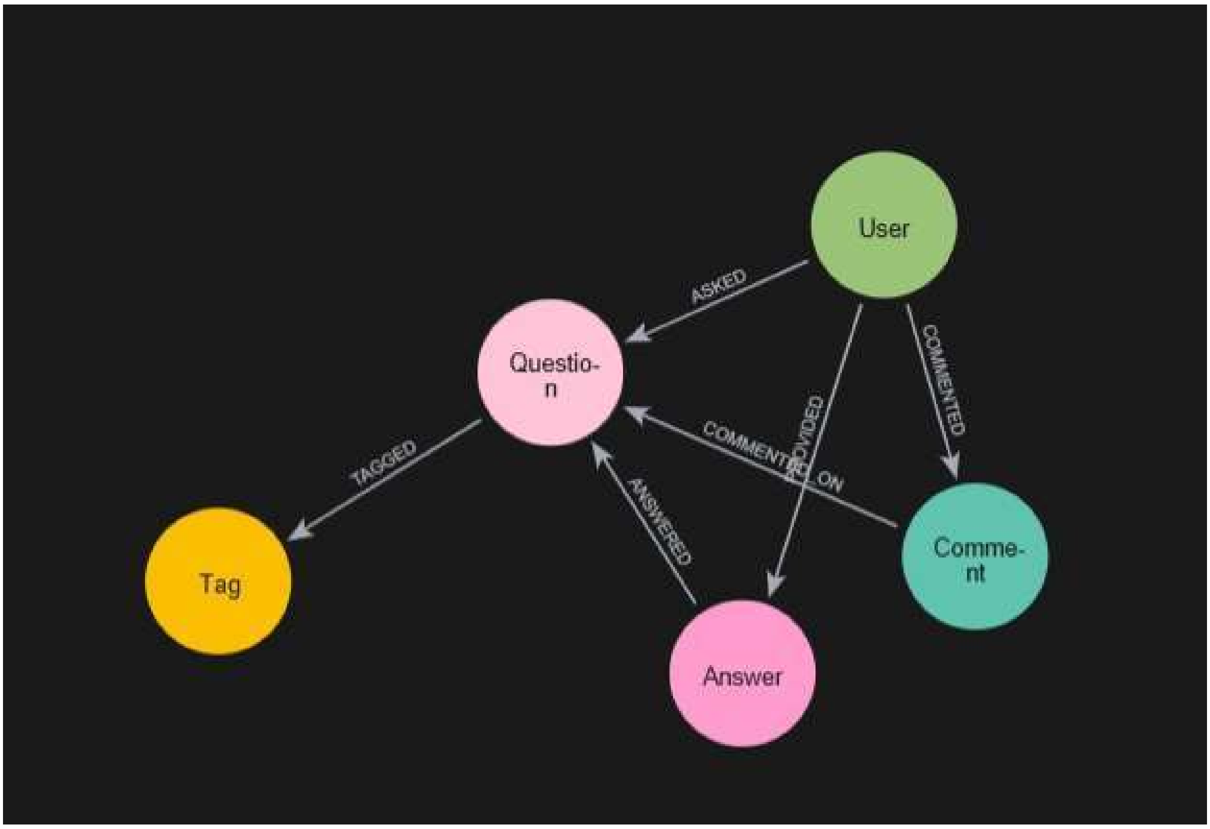
● Retrieve data from a social network API (e.g., Twitter, Facebook, or a custom dataset).

● Clean and preprocess the data to create a structured dataset suitable for Neo4j, including user profiles, posts, comments, and relationships.

● Perform analysis such as create follower following network,sentiment analysis,user behavior analysis.

Result: This experiment will help you gain a deep understanding of the social network's dynamics
and provide valuable insights into user behavior, content virality, community structures, and growth patterns. Additionally, it demonstrates the power of Neo4j as a graph database for social network analysis.

Display the schema db.schema.visualization();

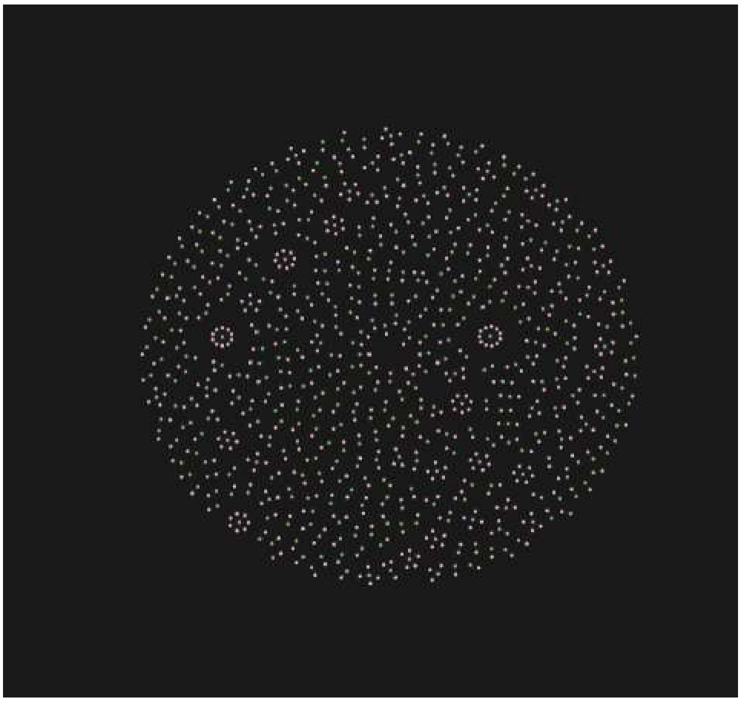


Map Questions and Users

(u:User)-[a:ASKED]->(q:Question)

u,a,q

MATCH RETURN

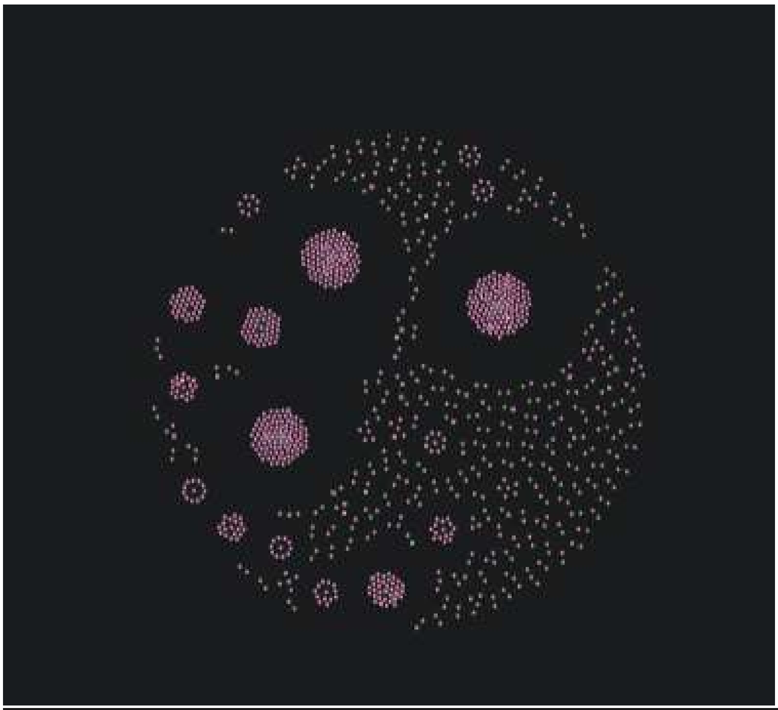


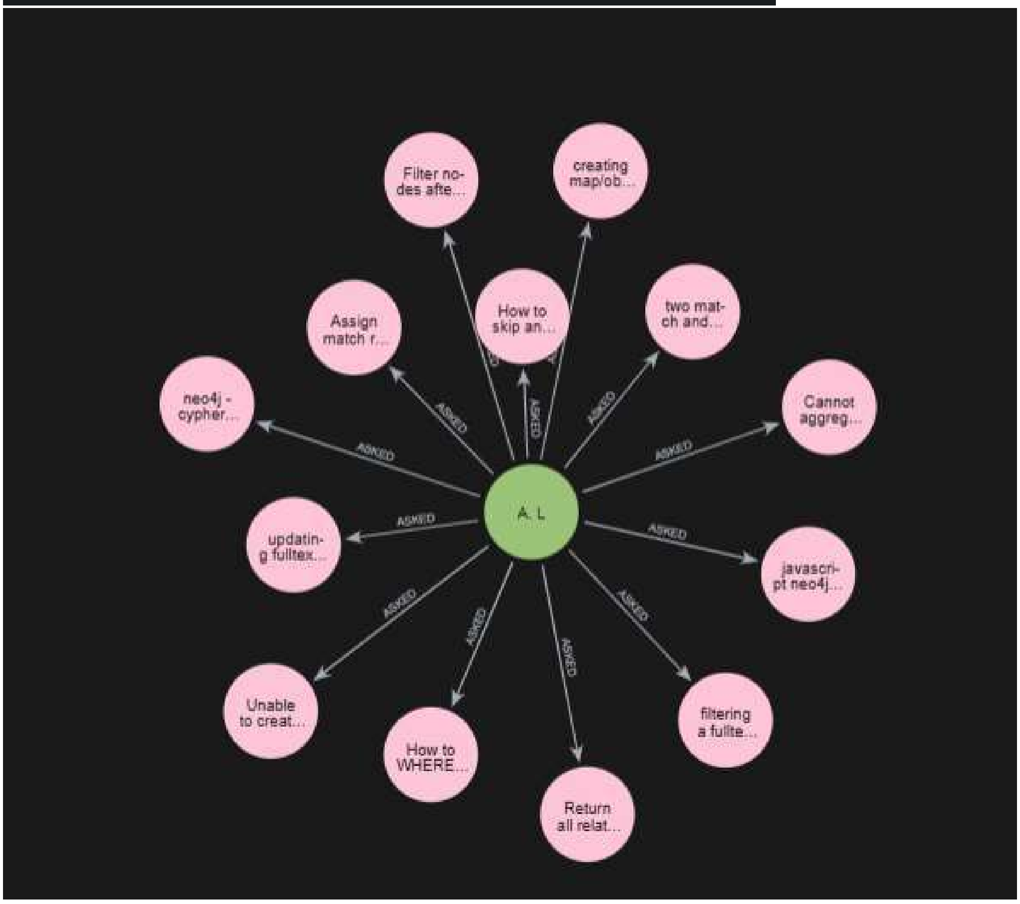
Map Questions Answered by Users

(u:User)-[p:PROVIDED]->(a:Answer)

[u,p,a

MATCH RETURN

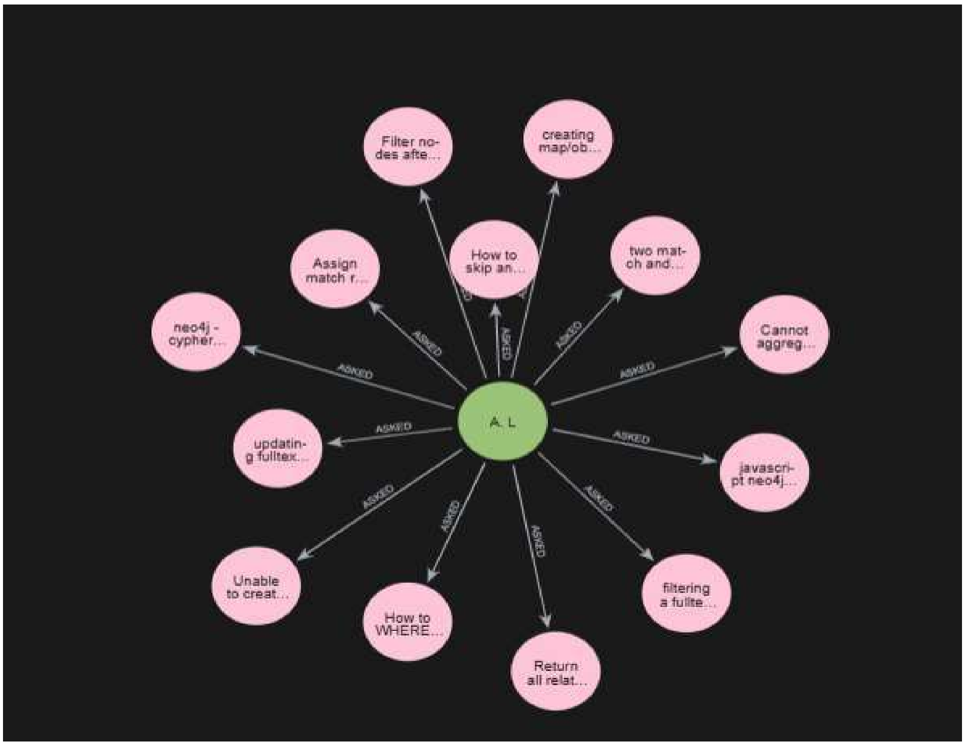




Using Where Clause

MATCH (u:UserHa:ASKEDl->(q:Question)

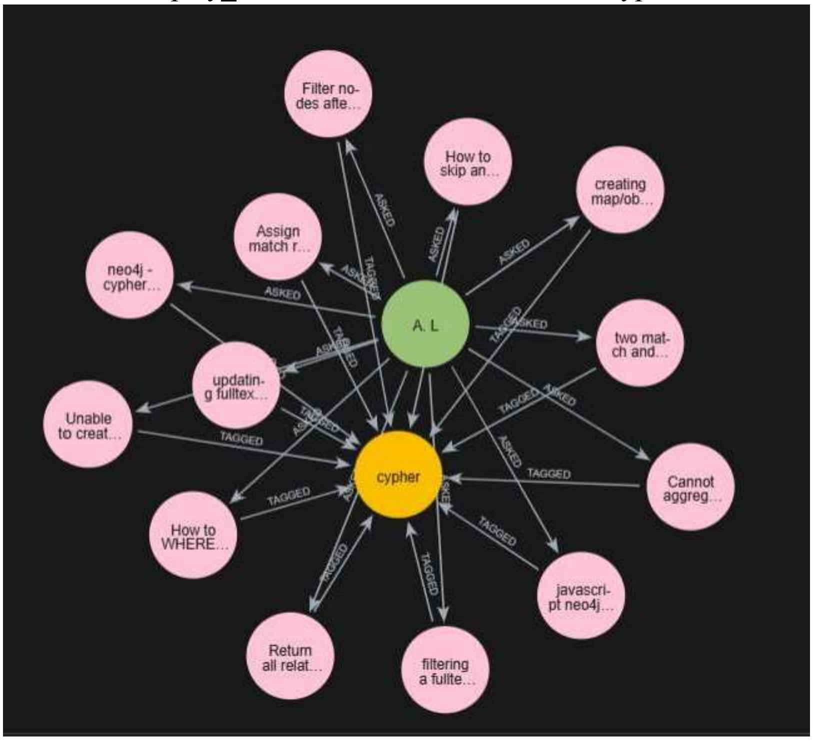
WHERE u.display_name = "A. L"



find all the questions asked by "A. L" also include tags

MATCHu:User)-[a:ASKED]->(q:Question)-[tg:TAGGED]->(t:Tag)

WHERE u.display_name = "A. L" AND t.name = "cypher"



Comments made by "cybersam" on questions tagged with "ne04j"

MATCH

(u:User)-[ct:COMMENTED]->(c:Comment)-[co:COMMENTED_ON]->(q:Question)-[tg:TAGGED]->(t:Tag)

WHERE u.display_name = "cybersam" AND t.name = "ne04j"

the shortest route between the user "A. L" who asked the most questions, and"cybersam" who provided the most answers.

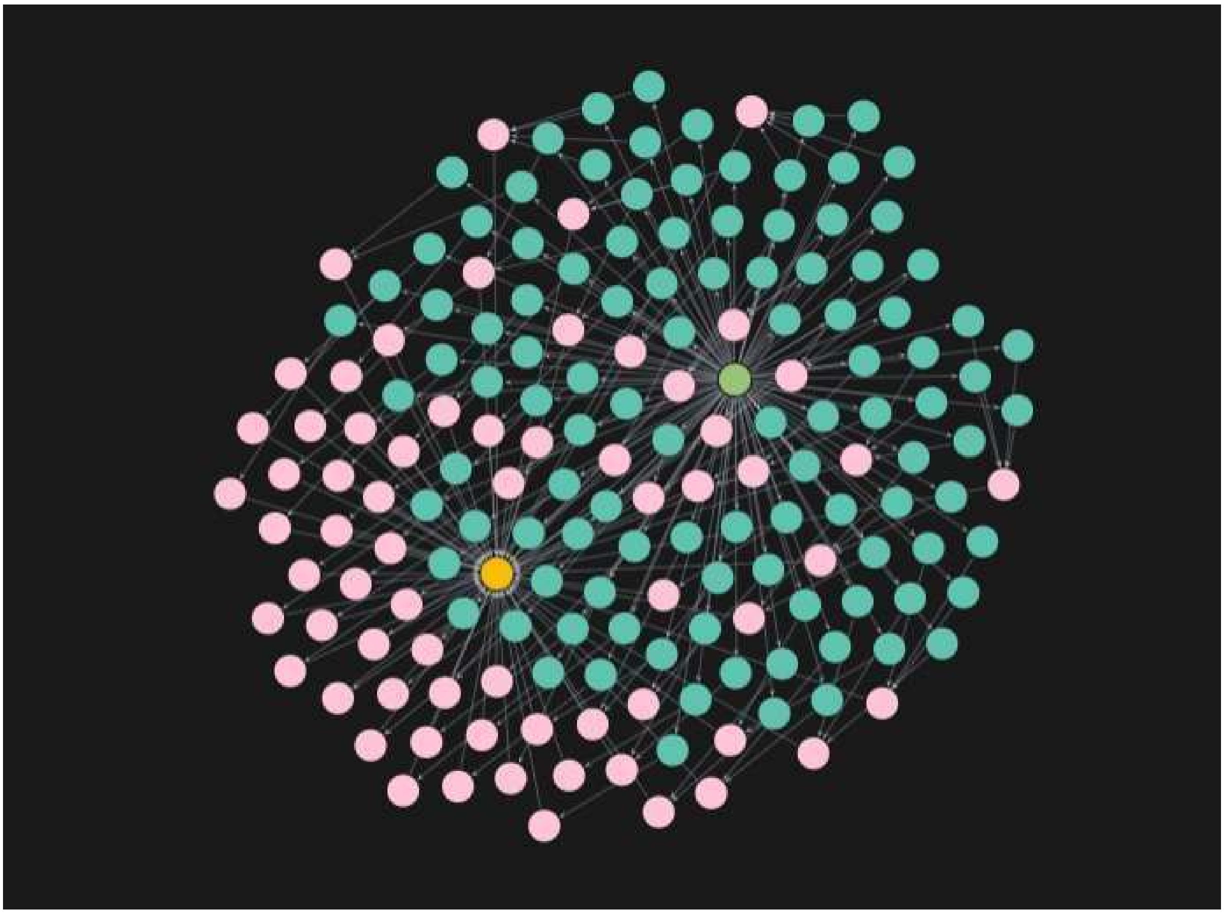
Comments made by "cybersam" on questions tagged with "ne04j"

MATCH

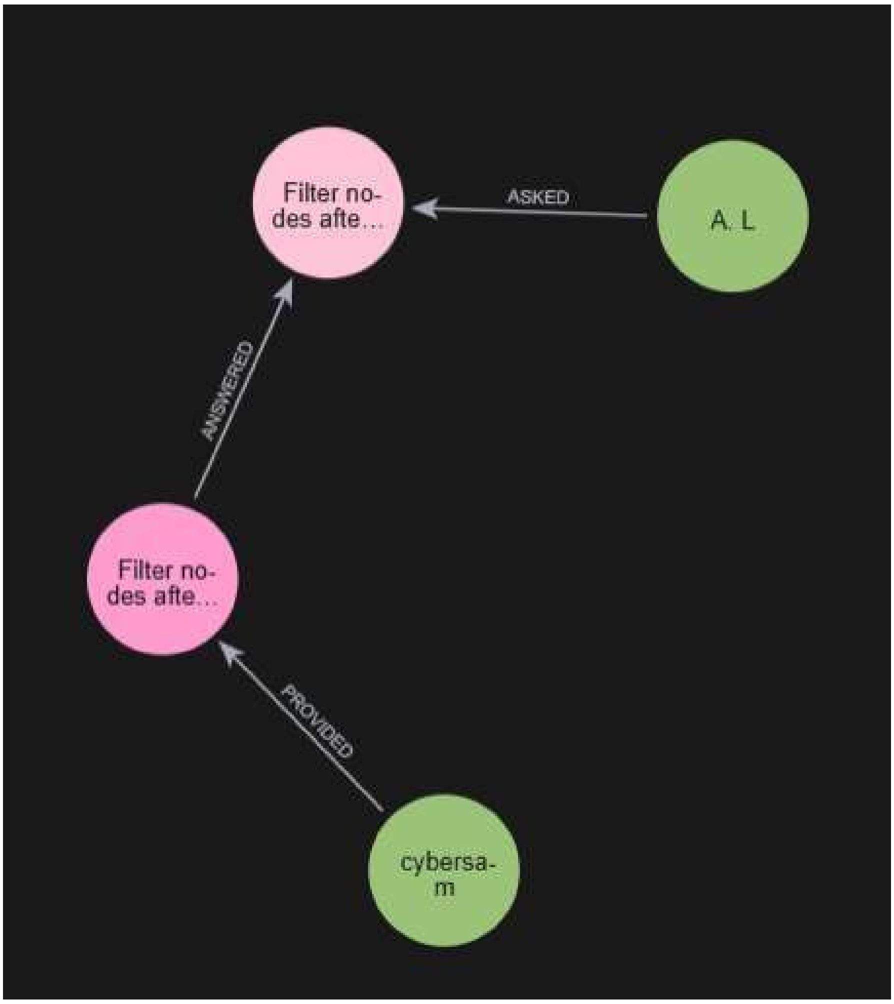
(u:User)-[ct:COMMENTED]->(c:Comment)-[co:COMMENTED_ON]->(q:Question)-[tg:TAGGED]->(t:Tag)

WHERE u.display_name = "cybersam" AND t.name = "ne04j"

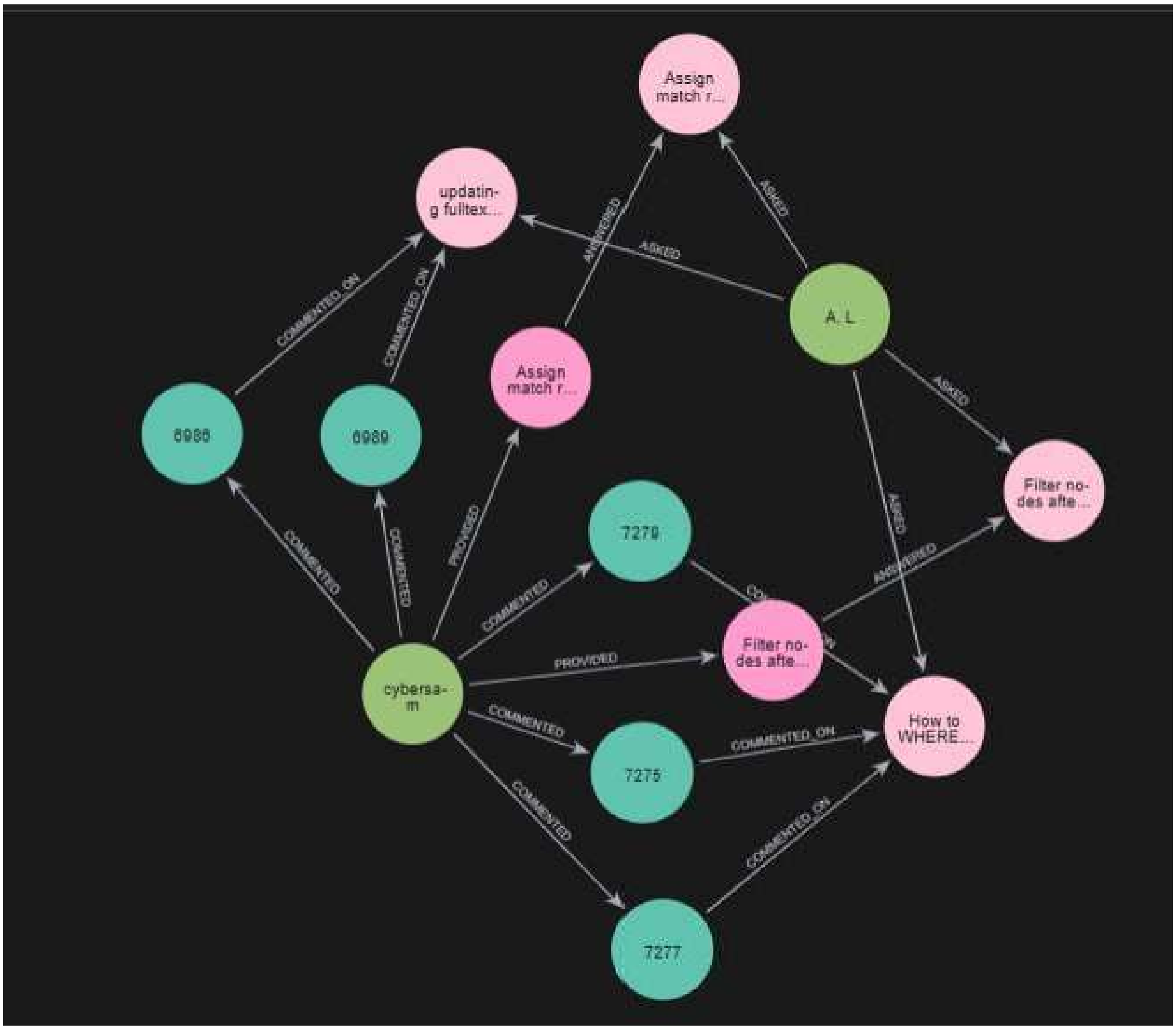
the shortest route between the user "A. L" who asked the most questions, and"cybersam" who provided the most answers.



MATCH path = shortestPath((ul:User {display_name: "A. L"})-[*]-(u2:User {display_name:"cybersam"}

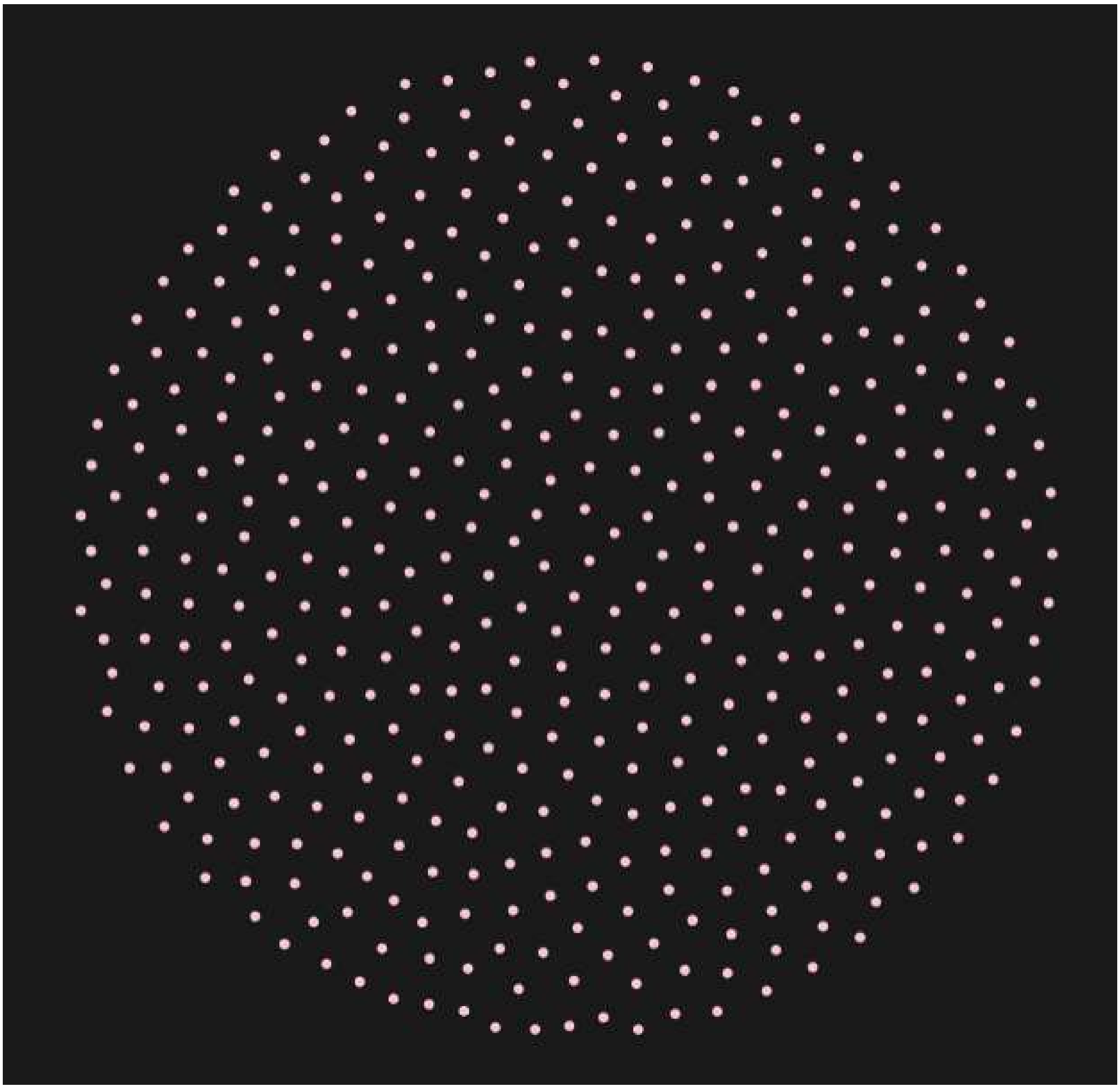


All the paths between the user "A. L" who asked the most questions, and "cybersam" who provided the most answers. MATCH path = allShortestPaths((ul :User {display_name: "A.L"})-[*]-(u2:User {display_name:"cybersam"})



What questions remain unanswered?

MATCH (q:Question)
WHERE NOT (q)<-[:ANSWERED]-()



Conclusion:

From this experiment, we can conclude that Neo4j is a powerful tool for analyzing social network dynamics. By representing users, posts, and interactions as a graph, we were able to visualize relationships, identify key influencers, and understand community structures effectively. The analysis provided insights into user behavior, content virality, and network growth patterns. Overall, this demonstrates how graph databases like Neo4j can uncover hidden patterns and valuable information in social networks that traditional relational databases may not capture efficiently.